Linear Algebra I – Week 6

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Reading and learning goals

Strang, *Introduction to Linear Algebra*, 4th ed., Sections 2.4–2.7, pp. 68–120 (Section 2.7 continues beyond p.119). Companion: *Linear Algebra and Its Applications*, 4th ed., Sections 1.4–1.6.

Topics: matrix rules; inverses and Gauss–Jordan; elimination matrices; LU, LDU; transposes, row exchanges and permutation matrices; applications.

Week 5 review: row and column pictures

Row picture: equations are intersecting lines or planes. Column picture: $Ax = x_1 a_1 + \cdots + x_n a_n = b$.

Example: $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ gives $(x, y) = (2, 1)$. Both pictures describe the same system.

Week 5 review: pivots

At pivot $a_{kk} \neq 0$, use multiplier $\ell_{ik} = a_{ik}/a_{kk}$ and $R_i \leftarrow R_i - \ell_{ik} R_k$.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$ Back substitution solves the triangular system.

Week 5 review: singular systems

$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has infinitely many solutions; replacing 6 by 7 gives the contradiction $0 = 1$.

For square matrices, a full set of pivots means a unique solution for every right-hand side.

Matrix multiplication: dimensions

For A of size $m \times n$ and B of size $n \times p$, AB has size $m \times p$ and $(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$.

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 4 \end{bmatrix}.$$

Four interpretations of AB

1. Entry: row i of A times column j of B.
2. Columns: $AB = [Ab_1 \ \cdots \ Ab_p]$.
3. Rows: each row of AB combines rows of B.
4. Outer products: $AB = \sum_k a_k r_k$, column k of A times row k of B.

Algebraic rules

$(AB)C = A(BC)$, $A(B + C) = AB + AC$, $(A + B)C = AC + BC$, $AI = IA = A$.

Generally $AB \neq BA$, and $(A + B)^2 = A^2 + AB + BA + B^2$, not $A^2 + 2AB + B^2$.

Noncommutativity: explicit example

$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$.

$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ but $BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$. Horizontal and vertical shears depend on the order applied.

Special matrices

I leaves vectors unchanged; diagonal $D = \mathrm{diag}(d_1, \ldots, d_n)$ scales coordinates.

Upper triangular matrices have zeros below the diagonal; lower triangular matrices have zeros above. Triangular systems admit forward/back substitution.

Elimination matrices

$R_2 \leftarrow R_2 - 3R_1$ corresponds to $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$.

$E \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$. The transformed system is $EAx = Eb$. Its inverse $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ restores the original rows.

Definition of the inverse

$AA^{-1} = A^{-1}A = I$. Then $Ax = b$ has solution $x = A^{-1}b$ for every b.

The inverse reverses the linear transformation. If $Ax = 0$ for a nonzero x, an inverse cannot exist.

Deriving the 2 by 2 inverse

$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = (ad - bc)I$. Therefore $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ when $ad - bc \neq 0$.

Example: $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.

Gauss–Jordan inversion

$[A \mid I] \to [I \mid A^{-1}]$ using identical row operations on both halves.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

Inverse columns and multiple solves

Write $A^{-1} = [x_1 \ \cdots \ x_n]$. Then $Ax_j = e_j$ for each standard basis vector e_j.

Finding the inverse means solving n systems. To solve just one $Ax = b$, triangular elimination is usually preferable to explicitly computing A^{-1}.

Inverse rules and proof

$(AB)^{-1} = B^{-1}A^{-1}$ because $(AB)(B^{-1}A^{-1}) = I$.

$(A^T)^{-1} = (A^{-1})^T$; $(A^{-1})^{-1} = A$; $(A^k)^{-1} = (A^{-1})^k$. The order of factors reverses.

Equivalent invertibility criteria

For square A: an inverse exists $\Leftrightarrow$ all pivots nonzero (after possible exchanges) $\Leftrightarrow$ columns independent $\Leftrightarrow$ $Ax = 0$ only for $x = 0$ $\Leftrightarrow$ unique solution to $Ax = b$ for every b.

Factorization $A = LU$

Elimination multipliers are stored below the diagonal in unit lower triangular L; elimination leaves upper triangular U.

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}}_{U}.$$ Check by multiplication.

Why elimination implies LU

If $E_k \cdots E_2 E_1 A = U$, then $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU$.

The inverse of subtracting a multiple of one row is adding that multiple back. The order is essential.

Detailed 3 by 3 factorization

$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 4 \\ -2 & 2 & 3 \end{bmatrix}$. Multipliers: $\ell_{21} = 2$, $\ell_{31} = -1$, $\ell_{32} = 1$.

$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$. Verify $LU = A$ row by row.

Solving $Ax = b$ through LU

With $A = LU$, solve $Ly = b$ (forward) and $Ux = y$ (backward).

For $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 19 \end{bmatrix}$, obtain $y = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ and $x = \begin{bmatrix} 3/2 \\ 2 \end{bmatrix}$. Factor once for many right sides.

From LU to LDU

Let D contain the nonzero diagonal pivots of U, and $\widehat{U} = D^{-1}U$. Then $A = LD\widehat{U}$, with unit diagonals in L and $\widehat{U}$.

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}.$$

Pivots, determinant and uniqueness

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $a \neq 0$, the pivots are a and $d - cb/a$; their product is $ad - bc = \det A$.

Without row exchanges, the determinant is the product of pivots. Normalized LDU is unique when it exists with nonzero pivots.

Why a row exchange is sometimes needed

$A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ is invertible but its first pivot is zero. Swap rows: $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$.

A zero pivot may signal poor row ordering rather than singularity.

Permutation matrices

Permutation matrices rearrange the rows of I; each row and column contains exactly one 1.

$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ exchanges rows: PA exchanges rows of A, whereas AP exchanges columns. There are $n!$ permutations of n rows.

Permutation transpose and inverse

Every permutation matrix satisfies $P^{-1} = P^T$ because $P^T P = I$.

$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ sends $(x_1, x_2, x_3)^T$ to $(x_2, x_3, x_1)^T$. It preserves vector lengths and inner products.

Pivoted factorization $PA = LU$

If exchanges are needed, collect them in a permutation matrix P so that $PA = LU$.

Example $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $L = I$, $U = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$. Pivoting also requires correct tracking of multipliers in L.

Partial pivoting: numerical motivation

$\begin{bmatrix} 10^{-8} & 1 \\ 1 & 1 \end{bmatrix}$ is nonsingular but using the tiny first pivot gives a multiplier 10^{8}. Exchanging the two rows avoids that large multiplier.

Partial pivoting selects the largest absolute entry below (and including) the current pivot in the active column.

Transposes and product order

$(A^T)_{ij} = a_{ji}$; $(A+B)^T = A^T + B^T$ and $(AB)^T = B^T A^T$.

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ has $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. A symmetric matrix satisfies $A^T = A$.

Symmetric LDL^T factorization

If symmetric A has an LDU factorization without exchanges, then its normalized upper factor is L^T: $A = LDL^T$.

$$\begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Application: difference and sum matrices

$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ produces $Ax = (x_1, x_2 - x_1, x_3 - x_2)^T$.

$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ produces cumulative sums. This is the discrete analogue of differentiation and integration.

Application: tridiagonal systems

$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ represents nearest-neighbor coupling or a finite-difference equation.

Its triangular factors preserve a narrow band. Solving with LU exploits this structure and avoids a dense inverse.

Takeaways and next week

1. Matrix multiplication has multiple equivalent interpretations but is not commutative.
2. An inverse exists only for nonsingular square matrices.
3. Elimination is encoded in $A = LU = LDU$.
4. Row exchanges are encoded by P and $PA = LU$.
5. Transposes and permutations prepare us for vector spaces and subspaces in Week 7.